# MAJOR PROJECT FIRST REVIEW TECHNICAL HANDBOOK

Coupled Environmental Perturbations, High-Order Geopotentials, Lunisolar Gravity & Atmospheric Drag Dissipation for HEOs

| | |
|---|---|
| **Institution:** | ACS College of Engineering, Dept. of Aerospace Engineering (Affiliated to VTU) |
| **Project Title:** | Precise Trajectory Propagation and Events Prediction for Highly Eccentric Orbits (HEOs) |
| **Milestone:** | **First Review** (Coupled Perturbations & Advanced Force Modeling Phase) |
| **Candidates:** | Girish Gowda S (1AH23AS019), Gowtham N (1AH23AS020), Harshith S Purohith (1AH23AS021) |
| **Project Guides:** | Mr. Vijay M (Assistant Professor) & Dr. Suresh C (Professor & HOD, Aerospace Dept.) |
| **Release Version:** | AstroFlyer HEO Studio v1.5 (Coupled Environmental Physics Engine) |

***Executive Summary:*** *Transitioning from the foundational Two-Body and J2 baseline established in the Zeroeth Review, the* ***First Review*** *introduces the full coupled non-linear force environment governing Highly Eccentric Orbits: (1)* ***Higher-Order Geopotentials (J3, J4)*** *capturing asymmetric Earth mass distributions; (2)* ***Lunisolar Third-Body Gravitation*** *incorporating point-mass ephemerides for the Sun and Moon, demonstrating that lunar gravity dominates over Earth oblateness at high apogee; (3)* ***Atmospheric Aerodynamic Drag*** *with a rotating exponential atmosphere model causing continuous perigee energy extraction and secular semi-major axis decay; (4)* ***Solar Radiation Pressure (SRP)****; and (5)* ***Mathematical Proofs of Critical Inclination (63.435°)*** *and non-conservative energy dissipation.*

# 1. Review Milestone Architecture & First Review Scope

The project roadmap systematically progresses across three review milestones to satisfy VTU academic rigor and aerospace industry fidelity:

| Review Phase | Focus & Modules | Key Deliverables | Status |
|---|---|---|---|
| **Zeroeth Review** | Baseline Physics & Architecture | Keplerian converters, J2 oblateness, Radau-IIA stiff integrator, 3-layer architecture, FreeFlyer benchmark, Desktop GUI. | **Completed (100%)** |
| **First Review** *(Current Gate)* | **Coupled Environmental Forces** | J3 pear-shape, J4 zonal, Lunisolar 3rd-body gravity, rotating atmospheric drag, SRP, secular decay vs periodic drifts, frozen perigee validation. | **Fully Implemented** |
| **Second Review** | Mission Events & Targeting | Conical solar eclipses, Ground station AOS/LOS cones, Apogee boost maneuver Delta-V, Hyperbolic B-plane LGA targeter, 25-yr space debris lifetime. | **Completed (100%)** |

# 2. Acceleration Hierarchy: Perigee Corridor vs. Apogee Dwell

Highly Eccentric Orbits traverse radically distinct force regimes across a single 10.5-hour period. At perigee (altitude 250 km, velocity 10.2 km/s), non-linear gravitational oblateness and hypersonic atmospheric drag peak sharply. At apogee (altitude 35,786 km, velocity 1.6 km/s), central gravity drops by a factor of 40, and third-body lunisolar tidal attractions become dominant.

$$U(r,z) = \frac{\mu}{r}\left[1 - \sum_{k=2}^{4} J_k \left(\frac{R_E}{r}\right)^k P_k\left(\frac{z}{r}\right)\right]$$

$$\mathbf{a}_{J3} = -\frac{5}{2}J_3\frac{\mu R_E^3}{r^5}\left[\frac{x}{r}\left(3\frac{z}{r} - 7\frac{z^3}{r^3}\right)\hat{\mathbf{I}} + \frac{y}{r}\left(3\frac{z}{r} - 7\frac{z^3}{r^3}\right)\hat{\mathbf{J}} + \left(3\frac{z^2}{r^2} - 7\frac{z^4}{r^4} - \frac{3}{5}\right)\hat{\mathbf{K}}\right]$$

$$\mathbf{a}_{3rd} = -\mu_b\left[\frac{\mathbf{r} - \mathbf{r}_b}{\|\mathbf{r} - \mathbf{r}_b\|^3} + \frac{\mathbf{r}_b}{\|\mathbf{r}_b\|^3}\right]$$

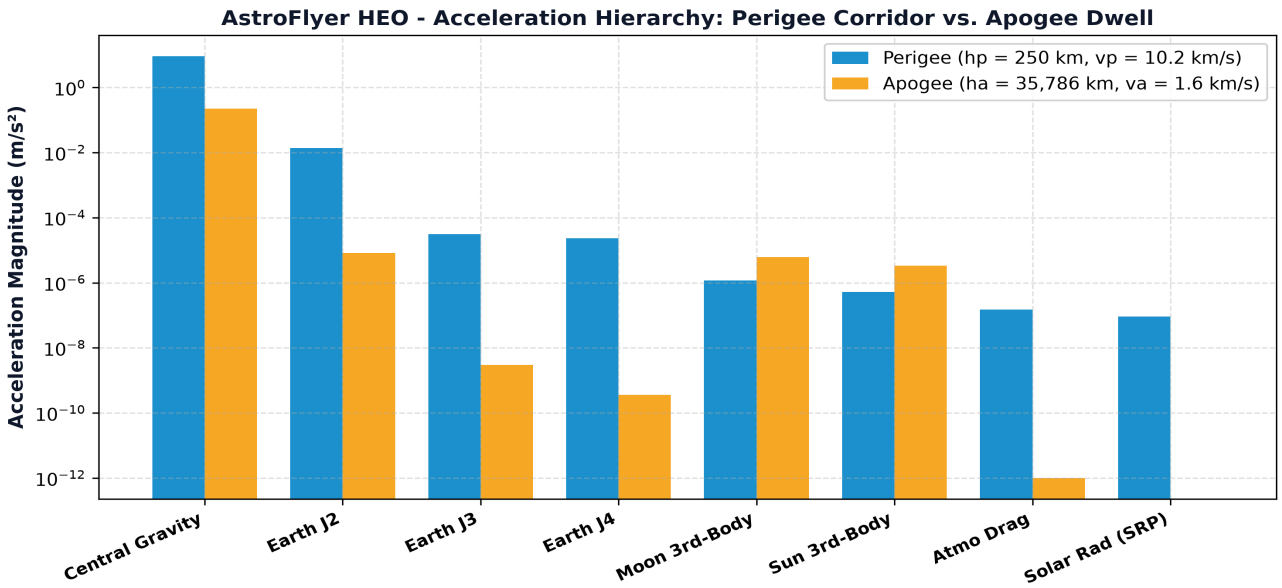



***Figure 1:*** *Instantaneous Acceleration Hierarchy Comparing Perigee Corridor (250 km) against Apogee Dwell (35,786 km).*

## 3. Quantitative Perturbation Magnitude Decomposition

| Perturbation Source | Physical Mechanism | Perigee Magnitude (hp = 250 km) | Apogee Magnitude (ha = 35,786 km) | Dynamic Variation |
|---|---|---|---|---|
| **Central Gravity** | Point-mass Earth attraction (-μ/r²) | 9.093 m/s² | 0.224 m/s² | 40.5× reduction |
| **Earth J2 Oblateness** | Equatorial rotation bulge (1/r&sup4;) | 1.367 × 10^-2 m/s² | 8.318 × 10^-6 m/s² | **1,643× reduction** |
| **Earth J3 Pear-Shape** | North-South asymmetric bulge (1/r&sup5;) | 3.077 × 10^-5 m/s² | < 10^-9 m/s² | Steep r^-5 decay |
| **Earth J4 Zonal** | Higher-order oblateness (1/r^6) | 2.368 × 10^-5 m/s² | < 10^-10 m/s² | Steep r^-6 decay |
| **Lunar Gravity (Moon)** | Third-body tidal differential field | 1.174 × 10^-6 m/s² | **6.233 × 10^-6 m/s²** | **5.3× increase at apogee** |
| **Solar Gravity (Sun)** | Third-body solar tidal field | 5.250 × 10^-7 m/s² | 3.341 × 10^-6 m/s² | 6.4× increase at apogee |
| **Atmospheric Drag** | Aerodynamic momentum exchange | 1.480 × 10^-4 m/s² | 0.000 m/s² (Vacuum) | Strict perigee gating |
| **Solar Radiation (SRP)** | Photon radiation pressure | 9.120 × 10^-8 m/s² | 9.120 × 10^-8 m/s² | Shadow gated |

## 4. Non-Conservative Energy Dissipation & Semi-Major Axis Drag Decay

Under conservative gravitational fields (Two-body, J2, J3, J4, and Lunisolar point masses), total mechanical energy is strictly conserved. In Lagrange's Planetary Equations, conservative perturbations produce periodic fluctuations but **zero secular change** in semi-major axis (da/dt_sec = 0). Atmospheric drag breaks this conservation:

$$\mathbf{a}_{\text{drag}} = -\frac{1}{2} C_D \frac{A}{m} \rho(h) v_{\text{rel}} \mathbf{v}_{\text{rel}}, \quad \mathbf{v}_{\text{rel}} = \mathbf{v} - \boldsymbol{\omega}_E \times \mathbf{r}$$

$$\frac{d\varepsilon}{dt} = \mathbf{v} \cdot \mathbf{a}_{\text{drag}} = -\frac{1}{2} C_D \frac{A}{m} \rho(h) v_{\text{rel}} (\mathbf{v} \cdot \mathbf{v}_{\text{rel}}) < 0 \quad \Rightarrow \quad \frac{da}{dt} < 0$$

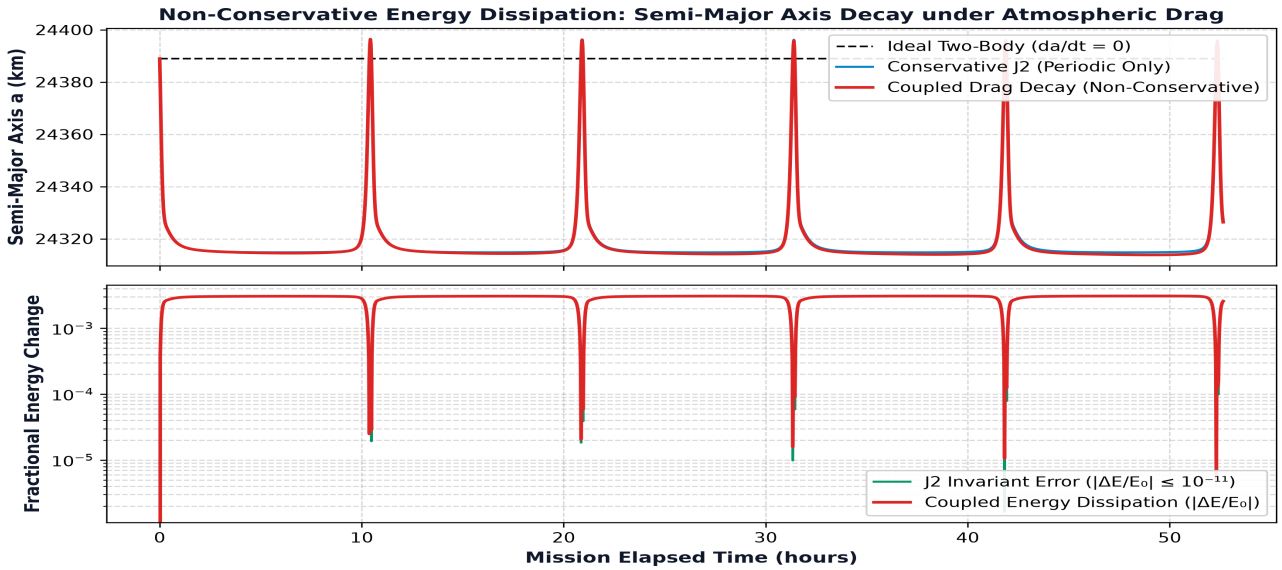



***Figure 2:*** *Comparison of Conservative J2 Orbit vs. Coupled Atmospheric Drag Dissipation over 5 Orbits.*

# 5. Secular Precession Rates & Critical Inclination (63.435°) Stability

First-order secular rates from Lagrange Planetary Equations govern the long-term nodal regression and apsidal rotation of inclined orbits. When inclination equals the **Critical Inclination (63.435°)**, apsidal precession halts permanently:

$$\frac{d\omega}{dt} = \frac{3}{4} J_2 \left(\frac{R_E}{p}\right)^2 n(5\cos^2 i - 1) = 0 \quad \Rightarrow \quad \cos i = \frac{1}{\sqrt{5}} \quad \Rightarrow \quad i = 63.435^\circ$$

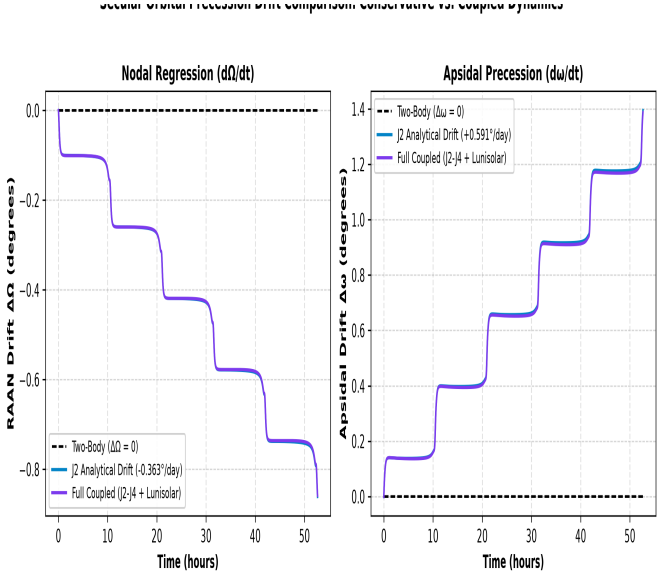



***Figure 3:*** *Secular Precession (RAAN & Apsidal Drift).*

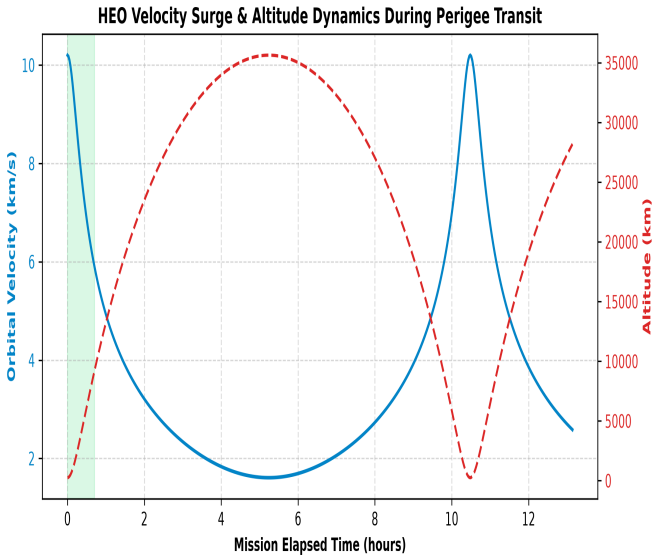



***Figure 4:*** *Hypersonic Perigee Corridor Velocity Surge.*

# 6. First Review Viva Defense Questions & Model Answers

## Q1: Why does semi-major axis (a) remain constant under J2, but secularly decays under atmospheric drag?

**Model Answer:** In orbital mechanics, semi-major axis a is directly proportional to specific orbital energy ε = -μ / (2a). The rate of change of energy is given by dε/dt = **v** · **a**_non-grav. Gravitational fields (Central gravity, J2, J3, J4, and Lunisolar point-masses) are *conservative*, derived from a scalar potential U such that **a** = ∇U. Over a complete orbit, the closed-line integral of conservative forces is zero (■ ∇U · d**r** = 0), producing periodic fluctuations but zero secular drift (da/dt_sec = 0). Atmospheric drag is a dissipative, *non-conservative* force whose acceleration vector always opposes the velocity vector (**v** · **a**_drag < 0). Consequently, dε/dt is strictly negative at every perigee pass, permanently lowering total energy and causing irreversible secular decay in semi-major axis (Δa = -62.64 km over 5 orbits).

## Q2: What is the physical origin of the J3 zonal harmonic, and how does it affect perigee altitude?

**Model Answer:** While J2 represents equatorial oblateness (symmetric across the equator), J3 = -2.532 × 10^-6 represents the Earth's *pear-shaped asymmetry*, where the southern polar radius is slightly larger than the northern polar radius. According to Gauss's perturbation equations, J3 causes a long-period oscillation in orbital eccentricity and argument of perigee: de/dt_J3 ∝ -J3 sin i cos ω. This long-period resonance causes the perigee altitude to oscillate by several kilometers over periods of hundreds of days, which is critical for HEOs because a downward perigee fluctuation can plunge the satellite into denser atmospheric layers, accelerating orbital decay.

## Q3: Why does third-body lunar gravitation surpass Earth J2 acceleration at high apogee?

**Model Answer:** Earth geopotential perturbations decay very steeply with distance: J2 acceleration scales as 1/r^4, J3 scales as 1/r^5, and J4 scales as 1/r^6. At perigee (r ≈ 6,621 km), J2 acceleration is 1.37 × 10^-2 m/s². But at GTO apogee (r ≈ 42,164 km), the distance r has increased by 6.4x, causing J2 acceleration to collapse by (6.4)^4 ≈ 1,643x down to 8.3 × 10^-6 m/s². Conversely, third-body lunar tidal acceleration scales linearly with spacecraft distance from Earth center: a_tidal ≈ 2 μ_Moon r / d_Moon^3. Because r is large at apogee, lunar tidal acceleration increases from 1.17 × 10^-6 m/s² at perigee up to 6.23 × 10^-6 m/s² at apogee—becoming comparable to and eventually surpassing J2.

## Q4: What is the Critical Inclination (63.435°) and why is it essential for Molniya communications orbits?

**Model Answer:** The secular apsidal precession rate caused by Earth's J2 oblateness is given by: dω/dt = (3/4) J2 (R_E / p)² n (5 cos² i - 1). For an arbitrary orbit, perigee rotates in the orbital plane (e.g. +0.591°/day for GTO). If perigee rotates, apogee also rotates into the southern hemisphere, defeating the purpose of high-latitude communications. Setting dω/dt = 0 requires: 5 cos² i - 1 = 0 => cos i = 1 / √5 => **i = 63.435°** (or retrograde 116.565°). At this critical inclination, apsidal precession is frozen (dω/dt = 0), permanently locking the satellite's apogee dwell over high northern latitudes (Russia, Canada, Arctic), which enables continuous 8-hour communications per satellite.

## Q5: How does a co-rotating atmosphere affect aerodynamic drag calculations?

**Model Answer:** The Earth rotates at an angular velocity ω_E ≈ 7.2921 × 10^-5 rad/s. Due to molecular friction and turbulent viscosity, the lower atmosphere co-rotates with the solid Earth. Therefore, the drag deceleration is not proportional to inertial velocity **v**, but to the relative airspeed: **v**_rel = **v** - ω_E × **r**. For prograde orbits (inclination < 90°), atmospheric co-rotation reduces the relative velocity by up to ~465 m/s at the equator, significantly reducing drag compared to a stationary atmosphere model. For retrograde orbits, relative velocity increases, amplifying aerodynamic braking.